THEME 1: CIVIC PULSE · DIGITAL GOVERNANCE & PUBLIC INFORMATION



CityPulse

Autonomous Multimodal AI Civic Triage & Closed-Loop Digital Governance

Making Every Civic Voice Count · Presenters: Dhrup, Soham, Jayvir

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PROBLEM STATEMENT

The Urban Governance Blindspot



Opaque Reporting

Citizens lack status or assignment visibility.



Manual Triage Bottlenecks

Manual triage can misroute issues and miss safety hazards.



Duplicate Sprawl

Duplicate reports waste administrative bandwidth.



Unverified Resolution

Unverified closure erodes public trust.

Source: Municipal grievance audit data · **Context:** Urban service delivery breakdown analysis

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AUTONOMOUS LIFECYCLE

CityPulse: Autonomous AI-Powered Civic Triage

A continuous, closed-loop pipeline transforming raw citizen submissions into verified infrastructure repairs.



Multimodal Intake

Capture a photo or description with precise location and municipal ward metadata.



Vision Classification

Gemini Flash vision classifies incident taxonomy and scores risk severity from 0 to 100.



Spatial Deduplication

Detect existing reports within a 150-meter radius, merging duplicates into community upvotes.



Closed-Loop Proof

Field crews submit repair evidence; citizens actively verify, rate quality, or dispute closure.

Core Mechanism: Gemini Flash multimodal inference paired with geospatial cluster heuristics

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ENTERPRISE FEATURES

Intelligent Capabilities Built for Municipal Scale

Engineered for departmental rigor, real-time spatial oversight, and direct citizen interaction.



Multimodal Civic Triage

Automated incident classification and priority scoring across six core sectors: roads, water, sanitation, lighting, drainage, and public safety.



Haversine Spatial Clustering

Geospatial clustering identifies recurring infrastructure hotspots, aggregating isolated tickets to justify systemic capital works.



Municipal Command Dashboard

Unified administrative operations console tracking ward-level performance, sector bottlenecks, team assignments, status, and real-time SLA countdowns.



Conversational Civic Assistant

Natural language citizen bot providing instant ticket status lookup, ward contact routing, and guidance on municipal policies and regulations.

Scale Scope: Municipal multi-ward deployment with unified departmental taxonomy

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TECHNICAL ARCHITECTURE

System Architecture

A decoupled, resilient stack combining multimodal AI reasoning with deterministic municipal offline fallback.



FastAPI + JWT

ASYNC REST API & AUTH

Gemini Flash

1.5 / 2.0 + HEURISTIC FALLBACK

PostgreSQL / SQLite

42 SAMPLE BENGALURU TICKETS

Stack: Vanilla JS SPA · Tailwind CSS · Chart.js · assistant bot · FastAPI (Pydantic v2) · Haversine Engine · Google Gemini Flash

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ACCOUNTABILITY PROTOCOL

Closing the Trust Deficit

Public confidence is restored when every stage of resolution is verified by physical evidence and clear community oversight.

Proof

Before & After Photo Verification

Mandatory photographic evidence uploaded by field crews before ticket closure.

24 / 48h

SLA Countdown and Escalation Target

Automated priority escalation alarms trigger as SLA deadlines approach.

Logs

Timestamped Audit Milestones

Immutable status trail from citizen submission to final verified sign-off.

Votes

Citizen Endorsement Elevators

Community upvoting signals collective urgency directly to ward administrators.

Governance Impact: Verified audit trails eliminate unconfirmed administrative ticket closures

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STRATEGIC HORIZON

The Future of Autonomous Smart Cities

Scaling from reactive civic complaint intake to predictive, proactive municipal infrastructure management.



Multichannel Access

Multilingual WhatsApp bot and regional voice IVR reporting to guarantee complete civic inclusion across every demographic.



Automated Auditing

Autonomous drone routes and municipal fleet dashcam computer vision to continuously scan roads and public fixtures.



Smart Escrow Contracts

Automated municipal contractor audits with quality-linked escrow disbursement released only upon citizen verification.



Predictive Maintenance

Early-warning machine learning models predicting storm drainage blockages and road structural degradation before monsoon events.

Roadmap: Proposed phased extensions for proactive municipal operating systems

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